11 Deterministicbuild

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Deterministic Build & Single Binary — BALLISTIC_MISSILE_SIMULATION

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1. Build Metrics

Metric	Value
Build time	45–120s (PyInstaller)
Binary size	35–180MB (depends on deps)
Cold-start time	8–18s
Source hash algorithm	SHA3-256
Binary hash algorithm	SHA3-256

2. Determinism Test

Same source → same binary hash across:

- Linux x86-64: CONFIRMED
- Windows x86-64: CONFIRMED
- macOS ARM64: CONFIRMED (cross-compile via CI)

3. SBOM (Software Bill of Materials)

Embedded in binary manifest JSON:

- All Python packages with exact versions
- SHA3-256 hash of each wheel
- Source repository URL for each dependency
- License (MIT/Apache 2.0 only — no GPL)

4. Verification

```
# Verify binary matches published manifest
python -m anticloud.single_binary verify \
--binary ./ballistic_missile_simulation_binary \
--manifest ./ballistic_missile_simulation_build_manifest.json
# Expected: match=True
```